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| United States Patent | 12389616        |
| Kind Code            | B2              |
| Date of Patent       | August 12, 2025 |
| Inventor(s)          | Gu; Man et al.  |

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### Transistors with multiple silicide layers

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#### Abstract

Structures for a transistor and methods of forming a structure for a transistor. The structure includes a first dielectric spacer, a second dielectric spacer, and a gate laterally between the first dielectric spacer and the second dielectric spacer. The gate includes a first silicide layer extending from the first dielectric spacer to the second dielectric spacer. The structure further includes a second silicide layer within the first silicide layer, and a contact that is aligned to the second silicide layer.

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**Inventors:** Gu; Man (Malta, NY), Yu; Hong (Clifton Park, NY), Peng; Jianwei (Clifton Park, NY), Wang; Haiting (Clifton Park, NY)

**Applicant:** GlobalFoundries U.S. Inc. (Malta, NY)

**Family ID:** 87430514

**Assignee:** GlobalFoundries U.S. Inc. (Malta, NY)

**Appl. No.:** 17/669584

**Filed:** February 11, 2022

#### Prior Publication Data

| Document Identifier | Publication Date |
|---------------------|------------------|
| US 20230261088 A1   | Aug. 17, 2023    |

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#### Publication Classification

**Int. Cl.:** H10D30/01 (20250101); H01L21/285 (20060101); H10D30/67 (20250101)

**U.S. Cl.:**

**CPC** H10D30/0213 (20250101); H01L21/28518 (20130101); H10D30/6735 (20250101); H10D30/6757 (20250101);

## Field of Classification Search

USPC: None

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*Primary Examiner:* Fairbanks; Brent A.

*Assistant Examiner:* Nielsen; Derek L

*Attorney, Agent or Firm:* Thompson Hine LLP

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## Background/Summary

### BACKGROUND

(1) The disclosure relates generally to semiconductor devices and integrated circuit fabrication and, in particular, to structures for a transistor and methods of forming a structure for a transistor.

(2) Complementary-metal-oxide-semiconductor (CMOS) processes may be employed to build a combination of p-type and n-type field-effect transistors that are used as devices to construct, for example, logic cells. Field-effect transistors generally include a source, a drain, a semiconductor body supplying a channel region between the source and drain, and a gate electrode overlapped with the semiconductor body. When a control voltage exceeding a characteristic threshold voltage is applied to the gate electrode, charge carrier flow occurs in the channel region between the source and drain to produce a device output current.

- (3) A silicide may be used as a contact material on the source, drain, and gate of a field-effect transistor. The silicide may be formed by a thermally-stimulated reaction between a metal or a metallic alloy thin film and the contacted semiconductor material. In conventional constructions for a field-effect transistor, the same silicide material is formed on the source, drain, and gate.
- (4) Improved structures for a transistor and methods of forming a structure for a transistor are needed.

## SUMMARY

- (5) In an embodiment, a structure for a transistor is provided. The structure includes a first dielectric spacer, a second dielectric spacer, and a gate laterally between the first dielectric spacer and the second dielectric spacer. The gate includes a first silicide layer extending from the first dielectric spacer to the second dielectric spacer. The structure further includes a second silicide layer within the first silicide layer, and a contact that is aligned to the second silicide layer.
- (6) In an embodiment, a structure for a transistor is provided. The structure includes a gate, and a raised source/drain region adjacent to the gate. The raised source/drain region comprises a semiconductor material. The structure further includes a silicide layer within the semiconductor material of the raised source/drain region, and a contact that is aligned to the silicide layer.
- (7) In an embodiment, a method of forming a structure for a transistor is provided. The method includes forming a gate positioned laterally between a first dielectric spacer and a second dielectric spacer. The gate includes a first silicide layer extending from the first dielectric spacer to the second dielectric spacer. The method further includes forming a second silicide layer within the first silicide layer, and forming a contact that is aligned to the second silicide layer.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various embodiments of the invention and, together with a general description of the invention given above and the detailed description of the embodiments given below, serve to explain the embodiments of the invention.
- (2) FIGS. 1-8 are cross-sectional views of a structure for a transistor at successive fabrication stages of a fabrication method in accordance with embodiments of the invention.
- (3) FIG. 9 is a cross-sectional view of a structure for a transistor in accordance with alternative embodiments of the invention

### DETAILED DESCRIPTION

- (4) With reference to FIG. 1 and in accordance with embodiments of the invention, a field-effect transistor **10** may be formed on a semiconductor layer **12**. A shallow trench isolation region **14** is formed by patterning a trench extending through the semiconductor layer **12** with lithography and etching processes, depositing a dielectric material to fill the trench, and planarizing and/or recessing the dielectric material. The shallow trench isolation region **14** may extend fully through the semiconductor layer **12** to a dielectric layer **16** and may fully surround a device region in which the field-effect transistor **10** is located. The shallow trench isolation region **14** and the dielectric layer **16** may each contain a dielectric material, such as silicon dioxide, that is an electrical insulator.
- (5) In an embodiment, the semiconductor layer **12** may be the device layer of a semiconductor-on-insulator substrate, which further includes a buried dielectric layer providing the dielectric layer **16** and a handle substrate **18**. In the representative embodiment, the semiconductor layer **12** may be comprised of a semiconductor material, such as single-crystal silicon. In an embodiment, the semiconductor layer **12** may have a thickness in a range of about 10 nanometers (nm) to about 200 nm.

(6) The field-effect transistor **10** may be formed by front-end-of-line processing as a device in the active region of the semiconductor layer **12**. The field-effect transistor **10** may include a gate **22** positioned above the semiconductor layer **12**, as well as a raised source/drain region **24** and a raised source/drain region **26** on respective sections of the semiconductor layer **12** adjacent to the gate **22**. The gate **22** is positioned in a lateral direction between the raised source/drain region **24** and the raised source/drain region **26**. Dielectric spacers **28**, **29** separate and electrically insulate the raised source/drain region **24** and the raised source/drain region **26** from the gate **22**. The dielectric spacers **28**, **29** may be comprised of a dielectric material, such as a low-k dielectric material.

(7) The gate **22** includes a semiconductor layer **21** comprised of a heavily-doped semiconductor material, such as polysilicon, that is deposited on the semiconductor layer **12** and then patterned with lithography and etching processes. The raised source/drain region **24** and the raised source/drain region **26** may be formed by epitaxial growth from respective sections of the semiconductor layer **12** adjacent to the opposite sidewalls of the gate **22**. The raised source/drain regions **24**, **26** may be epitaxially grown after the gate **22** is formed. The gate **22** has a thickness  $T1$  and a width  $W1$ , the raised source/drain region **24** has a thickness  $T2$  and a width  $W2$ , and the raised source/drain region **26** has a thickness  $T3$  and a width  $W3$ .

(8) The semiconductor material of the raised source/drain regions **24**, **26** may be doped (e.g., heavily doped) with a concentration of a dopant, such as an n-type dopant (e.g., phosphorus) that provides n-type conductivity. Alternatively, the semiconductor material of the raised source/drain regions **24**, **26** may be doped (e.g., heavily doped) with a concentration of a p-type dopant (e.g., boron) that provides p-type conductivity. Doped regions may be formed in the sections of the semiconductor layer **12** beneath the raised source/drain region **24**, **26** by dopant diffusion from the raised source/drain region **24**, **26**.

(9) As used herein, the term “source/drain region” means a doped region of semiconductor material that can function as either a source or a drain of a field-effect transistor. In an embodiment, the raised source/drain region **24** may be a raised source region of the field-effect transistor **10**, and the raised source/drain region **26** may be a raised drain region of the field-effect transistor **10**. In an alternative embodiment, the raised source/drain region **24** may be a raised drain region of the field-effect transistor **10**, and the raised source/drain region **26** may be a raised source region of the field-effect transistor **10**.

(10) A portion of the semiconductor layer **12** positioned beneath the gate **22** and laterally between the raised source/drain region **24** and the raised source/drain region **26** defines a channel region of the field-effect transistor **10**. The field-effect transistor **10** may include other elements such as a gate cap **30** comprised of a dielectric material, such as silicon nitride, that is positioned on the semiconductor layer **21** of the gate **22**, and a gate dielectric **32** that is positioned between the semiconductor layer **21** and the semiconductor layer **12**.

(11) With reference to FIG. **2** in which like reference numerals refer to like features in FIG. **1** and at a subsequent fabrication stage, a dielectric layer **33** is formed over the field-effect transistor **10**. The dielectric layer **33** may be comprised of a dielectric material, such as silicon dioxide, that is deposited and planarized by chemical-mechanical polishing to eliminate topography. The gate cap **30**, which may function as a polish stop during planarization, may be removed by a selective etching process to expose a top surface of the gate **22**. As used herein, the term “selective” in reference to a material removal process (e.g., etching) denotes that, with an appropriate etchant choice, the material removal rate (i.e., etch rate) for the targeted material is greater than the removal rate for at least another material exposed to the material removal process. The dielectric spacers **28**, **29** and dielectric layer **33** surround the space over the gate **22** opened by the removal of the gate cap **30**.

(12) After removing the gate cap **30**, the structure may be planarized by chemical-mechanical polishing to open the semiconductor layer **21** of the gate **22**. The top surface of the semiconductor layer **21** may operate as a polish stop during planarization. The planarization also shortens the

dielectric spacers **28**, **29** and thins the dielectric layer **33**.

(13) With reference to FIG. **3** in which like reference numerals refer to like features in FIG. **2** and at a subsequent fabrication stage, a layer **34** comprised of a silicide-forming metal is deposited as a coating on the semiconductor layer **21** of the gate **22**, the dielectric spacers **28**, **29**, and the dielectric layer **33**. The layer **34** may be in direct contact with the semiconductor layer **21** of the gate **22**, and may be subsequently used to silicide the gate **22**. In an embodiment, the silicide-forming metal contained in the layer **34** may be nickel included in a deposited coating of nickel-platinum. The layer **34** may be deposited by, for example, physical vapor deposition.

(14) With reference to FIG. **4** in which like reference numerals refer to like features in FIG. **3** and at a subsequent fabrication stage, a silicide layer **38** is formed from the semiconductor layer **21** of the gate **22** and the silicide-forming metal of the layer **34** that is in direct contact with the semiconductor layer **21**. The silicide layer **38** may be formed by a silicidation process that involves one or more annealing steps to form a silicide phase by reacting the silicide-forming metal in the deposited layer **34** with the semiconductor layer **21**. An initial annealing step of the silicidation process consumes the silicide-forming metal to form the silicide layer **38**. Following the initial annealing step, any remaining silicide-forming metal may be removed by wet chemical etching. The silicide layer **38** may then be subjected to an additional annealing step at a higher temperature to form a lower-resistance silicide phase.

(15) In an embodiment, the gate **22** may be partially silicided such that an upper portion of the gate **22** is comprised of the silicide layer **38** and a lower portion of the gate **22** is comprised of the semiconductor layer **21**. The silicide layer **38** constituting the upper portion of the gate **22** has a thickness **T3** that is less than the thickness **T1** of the gate **22**. In an embodiment, the silicide layer **38** may be comprised of nickel silicide. The silicide layer **38**, which has a width equal to the entire width **W1** of the gate **22**, spans fully across the gate **22** from the dielectric spacer **28** to the dielectric spacer **29**. The dielectric layer **33** blocks contact between the raised source/drain region **24**, **26** and the layer **34**, and therefore prevents silicidation of the raised source/drain region **24**, **26** during the formation of the silicide layer **38**.

(16) A semiconductor layer **36** comprised of a semiconductor material, such as silicon, may be formed on the silicide layer **38**. The semiconductor layer **36** may be used in a subsequent silicidation process, as subsequently described.

(17) With reference to FIG. **5** in which like reference numerals refer to like features in FIG. **4** and at a subsequent fabrication stage, the dielectric layer **33** may be etched back to reveal the raised source/drain regions **24**, **26**. A stress liner **40** is applied that coats the field-effect transistor **10**. The stress liner **40** may be comprised of a dielectric material, such as silicon nitride, deposited by plasma-enhanced chemical vapor deposition under deposition conditions that cause the dielectric material to be under stress. The deposition conditions may cause the stress liner **40** to incorporate tensile strain, which is transferred as tensile stress to the raised source/drain regions **24**, **26**. Alternatively, the deposition conditions may cause the stress liner **40** to incorporate compressive strain, which is transferred as compressive stress to the raised source/drain regions **24**, **26**.

(18) A dielectric layer **42** is formed over the stress liner **40**. The dielectric layer **42** may be comprised of a dielectric material, such as silicon dioxide, that is deposited and planarized by chemical-mechanical polishing to eliminate topography.

(19) With reference to FIG. **6** in which like reference numerals refer to like features in FIG. **5** and at a subsequent fabrication stage, contact openings **44**, **46**, **48** are patterned in the dielectric layer **42** and the stress liner **40**. The contact opening **44** penetrates through the dielectric layer **42** and the stress liner **40** to a portion of the raised source/drain region **24**. The contact opening **46** penetrates through the dielectric layer **42** and the stress liner **40** to a portion of the raised source/drain region **26**. The contact opening **48** penetrates through the dielectric layer **42** and the stress liner **40** to a portion of the semiconductor layer **36** positioned on the silicide layer **38** of the gate **22**.

(20) A layer **50** comprised of a silicide-forming metal is deposited inside the contact openings **44**,

**46, 48.** The layer **50** may be in direct contact with the semiconductor material of the raised source/drain region **24** over the area revealed at the bottom of the contact opening **44**, the semiconductor material of the raised source/drain region **26** over the area revealed at the bottom of the contact opening **46**, and the semiconductor layer **36** on the gate **22** over the area revealed at the bottom of the contact opening **48**. In an embodiment, the silicide-forming metal contained in the layer **50** may be titanium. The layer **50** may be deposited by, for example, physical vapor deposition.

(21) With reference to FIG. 7 in which like reference numerals refer to like features in FIG. 6 and at a subsequent fabrication stage, silicide layers **54, 56, 58** are formed using the silicide-forming metal of the layer **50**. The silicide layers **54, 56, 58** may be formed by a silicidation process that involves one or more annealing steps to create a silicide phase by reacting the silicide-forming metal of the deposited layer **50** at the respective bottoms of the contact openings **44, 46, 48** with the semiconductor materials of the raised source/drain region **24, 26** and the semiconductor layer **36** on the gate **22**. An initial annealing step of the silicidation process consumes the silicide-forming metal to form the silicide layers **54, 56, 58**. Following the initial annealing step, any remaining silicide-forming metal may be removed by wet chemical etching. Portions of the semiconductor layer **36** not consumed by the formation of the silicide layer **58** may be subsumed into the silicide layer **38** on the gate **22**. The silicide layers **54, 56, 58** may then be subjected to an additional annealing step at a higher temperature to form a lower-resistance silicide phase.

(22) The silicide layer **54** is embedded within the raised source/drain region **24** and positioned at the bottom of the contact opening **46**. The silicide layer **54** may be surrounded by and in direct contact with the semiconductor material of the raised source/drain region **24**. The contact opening **44** functions to position the silicide layer **54** relative to the raised source/drain region **24** and to limit the width of the silicide layer **54**. In particular, the width **W4** of the silicide layer **54** is less than the width **W1** of the raised source/drain region **24** such that the raised source/drain region **24** extends laterally beyond the silicide layer **54**. The silicide layer **54** extends to a depth within the raised source/drain region **24** that is less than the thickness **T2** of the raised source/drain region **24**. The silicide layer **54** is embedded in the raised source/drain regions **24** such that the silicide layer **54** is laterally positioned between different portions of the raised source/drain regions **24**, and another portion of the raised source/drain regions **24** is positioned in a vertical direction between the silicide layer **54** and the semiconductor layer **12**.

(23) The silicide layer **56** is embedded within the raised source/drain region **26** and positioned at the bottom of the contact opening **46**. The silicide layer **56** may be surrounded by and in direct contact with the semiconductor material of the raised source/drain region **26**. The contact opening **46** functions to position the silicide layer **56** relative to the raised source/drain region **26** and to limit the width of the silicide layer **56**. In particular, the width **W5** of the silicide layer is less than the width **W2** of the raised source/drain region **26** such that the raised source/drain region **26** extends laterally beyond the silicide layer **56**. The silicide layer **56** extends to a depth within the raised source/drain region **26** that is less than the thickness **T3** of the raised source/drain region **26**. The silicide layer **56** is embedded in the raised source/drain region **26** such that the silicide layer **56** is laterally positioned between different portions of the raised source/drain region **26**, and another portion of the raised source/drain region **26** is positioned in a vertical direction between the silicide layer **56** and the semiconductor layer **12**.

(24) The silicide layer **58** is embedded within the silicide layer **38** of the gate **22** and positioned at the bottom of the contact opening **48**. The silicide layer **58** may be surrounded by, and in direct contact with, the silicide layer **38** of the gate **22**. The contact opening **48** functions to position the silicide layer **58** relative to the silicide layer **38** and to limit the width of the silicide layer **58**. In particular, the width **W6** of the silicide layer **58** is less than the width **W1** of the silicide layer **38** and the width **W1** of the gate **22** such that the silicide layer **38** extends laterally beyond the silicide layer **58**. The silicide layer **58** extends to a depth within the silicide layer **38** of the gate **22** that is

less than the thickness of the silicide layer 38, which may be a fraction of the thickness T1 of the gate 22. The silicide layer 58 is embedded in the silicide layer 38 such that the silicide layer 58 is laterally positioned between different portions of the silicide layer 38, and another portion of the silicide layer 38 is positioned in a vertical direction between the silicide layer 58 and the semiconductor layer 21.

(25) With reference to FIG. 8 in which like reference numerals refer to like features in FIG. 7 and at a subsequent fabrication stage, a contact 64 is formed in the contact opening 44, a contact 66 is formed in the contact opening 46, and a contact 68 is formed in the contact opening 48. In an embodiment, the contacts 64, 66, 68 may be comprised of tungsten that is deposited by chemical vapor deposition and planarized.

(26) The contact 64 is coupled to the raised source/drain region 24 by the silicide layer 54. The contact 64 is in direct contact with the silicide layer 54 along an interface, and the contact 64 is aligned to the silicide layer 54 because both are formed with a width constraint at the interface imposed by the contact opening 44. In an embodiment, the width of the contact 64 at the interface may be equal to the width W4 of the silicide layer 54.

(27) The contact 66 is coupled to the raised source/drain region 26 by the silicide layer 56. The contact 66 is in direct contact with the silicide layer 56 along an interface, and the contact 66 is aligned to the silicide layer 56 because both are formed with a width constraint at the interface imposed by the contact opening 46. In an embodiment, the width of the contact 66 at the interface may be equal to the width W5 of the silicide layer 56.

(28) The contact 68 is coupled to the gate 22 by the silicide layer 58. The contact 68 is in direct contact with the silicide layer 58 along an interface, and the contact 68 is aligned to the silicide layer 58 because both are formed with a width constraint at the interface imposed by the contact opening 48. In an embodiment, the width of the contact 68 at the interface may be equal to the width W6 of the silicide layer 58.

(29) In contrast to conventional transistor constructions, the contacts 64, 66 are aligned to the silicide layers 54, 56 that are physically and electrically in contact with the raised source/drain regions 24, 26 and the contact 68 is aligned to the silicide layer 58 that is physically and electrically in contact with the gate 22. The dual silicide materials of the silicide layer 38 and the silicide layer 58 may be effective to reduce the gate resistance without the formation of the silicide layer 38 having any impact on the raised source/drain regions 24, 26. In that regard, the formation of the silicide layer 38 is independent of the formation of the silicide layers 54, 56, 58. The thickness of the raised source/drain regions 24, 26 may be reduced due to the decoupling provided by the separate formation of the silicide layer 38 of the gate 22 and the silicide layers 54, 56, 58.

(30) With reference to FIG. 9 in which like reference numerals refer to like features in FIG. 8 and in accordance with alternative embodiments, the gate 22 may be fully silicided such that the constituent semiconductor material is fully consumed during silicidation. As a result, the silicide layer 38 may extend fully in a vertical direction over the entire thickness T1 of the gate 22.

(31) The methods as described above are used in the fabrication of integrated circuit chips. The resulting integrated circuit chips can be distributed by the fabricator in raw wafer form (e.g., as a single wafer that has multiple unpackaged chips), as a bare die, or in a packaged form. The chip may be integrated with other chips, discrete circuit elements, and/or other signal processing devices as part of either an intermediate product or an end product. The end product can be any product that includes integrated circuit chips, such as computer products having a central processor or smartphones.

(32) References herein to terms modified by language of approximation, such as “about”, “approximately”, and “substantially”, are not to be limited to the precise value specified. The language of approximation may correspond to the precision of an instrument used to measure the value and, unless otherwise dependent on the precision of the instrument, may indicate a range of  $\pm 10\%$  of the stated value(s).

(33) References herein to terms such as “vertical”, “horizontal”, etc. are made by way of example, and not by way of limitation, to establish a frame of reference. The term “horizontal” as used herein is defined as a plane parallel to a conventional plane of a semiconductor substrate, regardless of its actual three-dimensional spatial orientation. The terms “vertical” and “normal” refer to a direction perpendicular to the horizontal, as just defined. The term “lateral” refers to a direction within the horizontal plane.

(34) A feature “connected” or “coupled” to or with another feature may be directly connected or coupled to or with the other feature or, instead, one or more intervening features may be present. A feature may be “directly connected” or “directly coupled” to or with another feature if intervening features are absent. A feature may be “indirectly connected” or “indirectly coupled” to or with another feature if at least one intervening feature is present. A feature “on” or “contacting” another feature may be directly on or in direct contact with the other feature or, instead, one or more intervening features may be present. A feature may be “directly on” or in “direct contact” with another feature if intervening features are absent. A feature may be “indirectly on” or in “indirect contact” with another feature if at least one intervening feature is present. Different features may “overlap” if a feature extends over, and covers a part of, another feature with either direct contact or indirect contact.

(35) The descriptions of the various embodiments of the present invention have been presented for purposes of illustration but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

## Claims

1. A structure for a transistor, the structure comprising: a first dielectric spacer; a second dielectric spacer; a gate laterally between the first dielectric spacer and the second dielectric spacer, the gate including a first silicide layer extending from the first dielectric spacer to the second dielectric spacer; a second silicide layer embedded within the first silicide layer, the second silicide layer extending to a depth in the first silicide layer, and the second silicide layer laterally positioned between a first portion of the first silicide layer and a second portion of the first silicide layer; and a first contact that is aligned to the second silicide layer.
2. The structure of claim 1 wherein the first contact directly contacts the second silicide layer at an interface, the first contact has a first width at the interface, the second silicide layer has a second width at the interface, and the first width is equal to the second width.
3. The structure of claim 2 wherein the first silicide layer has a third width greater than the second width.
4. The structure of claim 2 wherein the first silicide layer has a first thickness, and the second silicide layer has a second thickness that is less than the first thickness.
5. The structure of claim 1 wherein the gate further includes a semiconductor layer extending from the first dielectric spacer to the second dielectric spacer, and the first silicide layer is stacked with the semiconductor layer.
6. The structure of claim 1 wherein the gate fully comprises the first silicide layer.
7. The structure of claim 1 further comprising: a raised source/drain region comprising a semiconductor material, the raised source/drain region positioned adjacent to the gate; a third silicide layer within the semiconductor material of the raised source/drain region; and a second contact that is aligned to the third silicide layer.
8. The structure of claim 7 wherein the third silicide layer has a first width, and the semiconductor



material of the raised source/drain region has a second width that is greater than the first width.

9. The structure of claim 8 wherein the semiconductor material of the raised source/drain region has a first portion and a second portion, and the third silicide layer is laterally positioned between the first portion of the raised source/drain region and the second portion of the raised source/drain region.

10. The structure of claim 1 wherein the first silicide layer comprises a first silicide material, and the second silicide layer comprises a second silicide material having a different composition than the first silicide material.

11. The structure of claim 10 wherein the first silicide material is nickel silicide, and the second silicide material is titanium silicide.

12. The structure of claim 1 further comprising: a silicon-on-insulator substrate including a buried dielectric layer and a device layer on the buried dielectric layer, wherein the gate is positioned above the device layer.

13. The structure of claim 1 further comprising: a stress liner over the gate; a dielectric layer on the stress liner; and a contact inside a contact opening extending through the stress liner and the dielectric layer, wherein the second silicide layer is positioned inside the contact opening between the contact and the first silicide layer.

14. The structure of claim 13 wherein the first portion of the first silicide layer is laterally positioned between the second silicide layer and the first dielectric spacer, the second portion of the first silicide layer is laterally positioned between the second silicide layer and the second dielectric spacer, and the first contact includes a portion laterally positioned between the first portion of the first silicide layer and the second portion of the first silicide layer.

15. The structure of claim 1 wherein the first portion of the first silicide layer is laterally positioned between the second silicide layer and the first dielectric spacer, the second portion of the first silicide layer is laterally positioned between the second silicide layer and the second dielectric spacer, and the first contact includes a portion laterally positioned between the first portion of the first silicide layer and the second portion of the first silicide layer.

16. The structure of claim 1 wherein the first silicide layer has a thickness, and the depth of the second silicide layer is less than the thickness.

17. A method of forming a structure for a transistor, the method comprising: forming a gate, wherein the gate is positioned laterally between a first dielectric spacer and a second dielectric spacer, and the gate includes a first silicide layer extending from the first dielectric spacer to the second dielectric spacer; forming a second silicide layer embedded within the first silicide layer, wherein the second silicide layer extends to a depth in the first silicide layer, and the second silicide layer is laterally positioned between a first portion of the first silicide layer and a second portion of the first silicide layer; and forming a first contact that is aligned to the second silicide layer.

18. The method of claim 17 further comprising: forming a raised source/drain region comprising semiconductor material, wherein the raised source/drain region is positioned adjacent to the gate; forming a third silicide layer within the semiconductor material of the raised source/drain region; and forming a second contact aligned to the third silicide layer.

19. The method of claim 17 wherein the first contact directly contacts the second silicide layer at an interface, the first contact has a first width at the interface, the second silicide layer has a second width at the interface, and the first width is equal to the second width.

20. The method of claim 19 wherein the first silicide layer has a third width greater than the second width, the first silicide layer has a first thickness, and the second silicide layer has a second thickness that is less than the first thickness.

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